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Plumeria rubra

Plumeria rubra is a deciduous plant species belonging to the genus *Plumeria*.[4] Originally native to Mexico, Central America, Colombia and Venezuela, it has been widely cultivated in subtropical and tropical climates worldwide and is a popular garden and park plant, as well as being used in temples and cemeteries. It grows as a spreading tree to 7–8 m (23–26 ft) high and wide, and is flushed with fragrant flowers of shades of pink, white and yellow over the summer and autumn.

Its common names include **frangipani**,[2] **red paucipan**,[2] **red-jasmine**,[2] **red frangipani**, **common frangipani, temple tree**,[2] **calachuchi**,[5] or simply **plumeria**. Despite its common name, the species is not a "true jasmine" and not of the genus *Jasminum*.

Taxonomy

Plumeria rubra was one of the many species first described by Carl Linnaeus, and appeared in the 1753 edition of *Species Plantarum*. Its specific epithet is derived from the Latin *ruber* "red". The epithets *acuminata*, *acutifolia*, and *lutea* are seen, but these are invalid.

Common names

The common name "frangipani" comes from the Italian Frangipani family, a sixteenth-century marquess of which invented a plumeria-scented perfume. The genus name honors Charles Plumier, who was a French monk of the Franciscan order, and a botanist.[6]

In its native range in Mexico the common name is *cacaloxochitl* or *cacaloxuchitl*. The name comes from Nahuatl and means "crow's flower".[7] *P. rubra* was declared the national flower of Nicaragua in 1971, where it is known as *sacuanjoche*.[8] In Spanish, frangipanis are also referred to as *alhelí*, *alhelí cimarrón*, and *suche*.[9]

P. rubra entered Southeast Asia via the Manila galleons from Mexico to the Philippines in the 1560s. It retained the Nahuatl-derived name in the Philippine languages where it is known as *kalachuchi* (also spelled *kalatsutsi*), or calachuchi in Philippine

Plumeria rubra

Tree with pink flowers in Pakistan

Conservation status

Least Concern (IUCN 3.1)[1]

Scientific classification

Kingdom:	Plantae
Clade:	Tracheophytes
Clade:	Angiosperms
Clade:	Eudicots
Clade:	Asterids
Order:	Gentianales
Family:	Apocynaceae
Genus:	*Plumeria*
Species:	***P. rubra***

Binomial name

Plumeria rubra
L.[2]

Synonyms[3]

List

- *Plumeria acuminata* W.T.Aiton
- *Plumeria acutifolia* Poir.
- *Plumeria angustifolia* A.DC.
- *Plumeria arborea* Noronha

English.[5][10] The Nahuatl name is also partly preserved in the Ternate common name *tsjutsju*, derived from the Philippine name.[10] In other parts of Indonesia and Malaysia, it acquired different names like *cempaka* or *chempaka* derived from the resemblance of its fragrance to the champaca tree. It also acquired the name *bunga kamboja* (literally "flower of Cambodia") in Malay, referencing its foreign origin. Though it is not native to Cambodia nor is there evidence that it spread from Cambodia. Due to its associations with graveyards and death, it is also known as *bunga kuburan*, literally "grave flower".[10][11]

In the Pacific Islands where it was introduced in the late 19th century,[10] *P. rubra* is known as *melia* in Hawaii,[12] and *tipani* in the Cook Islands.[13]

Elsewhere, it is known as *champa* (or a derivative thereof) in India and Pakistan.[13] It is known by many names in Brazil, including *jasmim-de-caiena*, *jasmin-do-pará*, and *jasmin-manga*.[9] In Portuguese, it is also known as *flor-de-Saint-Antunio*.[9] In Cambodia it is given the names *châmpéi krahâ:m* (also romanised as *krahom*, meaning 'red'), or *châmpéi slük sruëch*, while the French term for the species is *frangipanier à fleurs rouges*.[14] In Sri Lanka, it is known as *araliya* in Sinhalese. In Myanmar, it is referred to as *mawk-sam-ka*, *mawk-sam-pailong*, and *sonpabataing*.[15] In China, it has the common name *ji dan hua*, and in the United States, it is referred to as *nosegay*.[13]

Description

Plumeria rubra belongs to the dogbane family and grows as a spreading shrub or small tree to a height of 2–8 m (5–25 ft) and similar width.[6] It has a thick succulent trunk and sausage-like blunt branches covered with a thin grey bark. The branches are somewhat brittle and when broken, ooze a white latex that can be irritating to the skin and mucous membranes. This latex found in the stem of the plants is in fact toxic, but not deadly unless present in large quantities.[16] The large green leaves can reach 30 to 50 cm (12 to 20 in) long and are arranged alternately and clustered at the end of the branches. The boles of these plants can be up to 25 cm in the wild. It tends to be smaller in cultivation.[16]

They are deciduous, falling in the cooler months of the year. The flowers are terminal, appearing at the ends of branches over the summer. Often profuse and very prominent, they are strongly fragrant, and have five petals. The flowers give off their

- *Plumeria arborescens* G.Don
- *Plumeria aurantia* Lodd. ex G.Don
- *Plumeria aurantia* Endl.
- *Plumeria aurantiaca* Steud.
- *Plumeria bicolor* Ruiz & Pav.
- *Plumeria carinata* Ruiz & Pav.
- *Plumeria conspicua* G.Don
- *Plumeria gouanii* D.Don ex G.Don
- *Plumeria incarnata* Mill.
- *Plumeria incarnata* var. *milleri* (G.Don) A.DC.
- *Plumeria jamesonii* Hook.
- *Plumeria kerrii* G.Don
- *Plumeria kunthiana* Kostel.
- *Plumeria lambertiana* Lindl.
- *Plumeria loranthifolia* Müll.Arg.
- *Plumeria lutea* Ruiz & Pav.
- *Plumeria mariaelenae* J.F.Gut. & J.Linares
- *Plumeria megaphylla* A.DC.
- *Plumeria mexicana* G.Lodd.
- *Plumeria milleri* G.Don
- *Plumeria mollis* Kunth
- *Plumeria northiana* Lodd. ex G.Don
- *Plumeria purpurea* Ruiz & Pav.
- *Plumeria rubra* var. *acuminata* (W.T.Aiton) R.S.Rao & Balamani
- *Plumeria rubra* f. *acutifolia* (Poir.) Woodson
- *Plumeria rubra* f. *lutea* (Ruiz & Pav.) Woodson
- *Plumeria rubra* f. *tricolor* (Ruiz

fragrance in the morning and in the evening. This fragrance is similar to that of rose, citrus, and cinnamon.[17] The colors range from the common pink to white with shades of yellow in the centre of the flower.[4][18] Initially tubular before opening out, the flowers are 5–7.5 cm (2–3 in) in diameter, and only rarely go on to produce seed - 20-60 winged seeds are contained in a 17.5 cm (7 in) pod.[12] The fruits are cylindrical pods that are rarely found in cultivation.[6]

& Pav.) Woodson

- *Plumeria rubra* f. *typica* Woodson
- *Plumeria tenuifolia* Lodd. ex G.Don
- *Plumeria tricolor* Ruiz & Pav.

Distribution and habitat

Its native range extends from Baja California into central Mexico south through Central America (Costa Rica, Nicaragua, Guatemala, El Salvador, Belize, Honduras, Panama) to the Department of Antioquia in Colombia and Venezuela in South America.[13] It is cultivated widely in tropical and subtropical regions of the world, and has become naturalized in most parts of tropical Asia and the Pacific Islands.[19][2][20]

Flower of *Plumeria rubra* in West Bengal, India.

P. rubra was introduced in the 1560s into the Philippines during the Spanish colonial period. From there, it spread to the rest of Southeast Asia and into other parts of the Asia. It was also introduced into the Pacific Islands in the late 19th century, where it has since become iconic due to their use in leis.[10]

P. rubra generally inhabits hot and rocky areas with dry to moderate rainfall. They can survive in locations with prominent dry seasons, where they can flower on the bare branches, or in more humid conditions, where they can remain evergreen. It can also be found in rocky forests, mountain slopes, and even occasionally on plains or savannas. It occupies elevations of 500 to 1000 meters but can be found up to elevations of 1500 meters.[16]

Cultivation

The species is cultivated around the world in subtropical and tropical climates. In Australia, it is widely seen in cultivation in Sydney and Perth and warmer frost free climates northwards.[21] In the mainland United States, it tolerates USDA Hardiness zones 10B to 11 (southern coastal California and the southern tip of Florida).[18] It is also grown in Hawaii to an altitude of 2000 m.[12] They tolerate a wide variety of soils, from acid to alkaline and sandy to clay.[18] These plants grow best in dry to medium moisture, well-drained soils in full sun and will bloom through most of the year in tropical areas. They do not grow well in wet soils and in areas with temperatures below 10 °C (50 °F) during the winter seasons, the plants will stop blooming and shed their leaves.[6] Established plants are also very salt tolerant

A flowering *Plumeria rubra* in Piracicaba, Brazil.

and tolerate even salt-laden winds.[16] Widely available in nurseries, frangipanis are readily propagated by cuttings of branches taken in cooler months and left to dry for a week or more.[21] As well as gardens and street- and park planting, frangipanis are planted in temples and cemeteries.[12]

Plumeria rubra is an important crop in Hawaii, with over 14 million flowers sold to be used in leis there in 2005.[22]

In temperate areas *P. rubra* must be grown under glass, in a large conservatory or similar, due to its requirement for warm conditions. However it may be placed outside in a sheltered sunny spot during the summer months. In the United Kingdom it has gained the Royal Horticultural Society's Award of Garden Merit.[23][24]

Some forms in cultivation are hybrids between this species and *Plumeria obtusa*; these have rounded rather than pointed leaves and are less likely to be deciduous.[12] The white and yellow flowered “Singapore" cultivar, also from the United States, usually holds it’s leaves all year round in Hawaii.[25]

Coleosporium plumeriae, known as plumeria rust or frangipani rust, is a fungus which attacks young leaves of *P. rubra*. It causes a brownish or orange powdery coating or blistering of leaves. It has been recorded from Hawaii and the east coast of Australia.[12][26] It was first recognised by the French mycologist Narcisse Théophile Patouillard on Guadeloupe Island in the eastern Caribbean in 1902, and had reached Taiwan by 2005.[27]

Uses

The USDA Forestry Service lists *Plumeria rubra* as a poisonous plant and warns against touching or eating any part of the plant.[28]

In Cambodia, as with other *Plumeria* species, *P. rubra* flowers is used for necklaces, as offerings to deities or as decoration for coffins. The leaves of this species are used in the care of sores and made into soothing infusions.[14] The flowers and bark of it are also used in traditional Chinese medicine in the treatment of fever, bacillary dysentery, pertussis and so on.[29] Chinese herbal teas such as Wang Lao Ji also contain P. rubra.

P. rubra possesses fulvoplumierin, an antibiotic that inhibits the growth of *Mycobacterium tuberculosis*. The plant has also been shown to be an antifungal, antiviral, analgesic, antispasmodic, and hypoglycemic.[16] *P. rubra* is also reported to contain agoniadin, plumierid, plumeric acid, cerotinic acid, and lupeol, and the stem possesses an alkaloid called triterpinoid. The plant has been known to promote digestion and excretion, along with respiratory and immune functions. The sap of the plant is used as a laxative and is a remedy for bloating and stomachaches. The bark is said to be purgative and is also used for venereal sores. The flowers can be boiled in water or juice and made into a salad to promote bowel movement, urine flow, and to control gas and phlegm. The flowers are also used in the treatment of asthma.[15]

On the Molokai island in the Hawaiian archipelago, *P. rubra* is cultivated in order to produce neck garlands (leis).[17] They are also used to make a scented oil in many Pacific islands that include Hawaii.[6] The flowers are used to scent coconut oil. The bark contains faint purplish streaks and the wood is hard and compact with a very fine texture. The wood takes a high polish.[16]

As a symbol

Plumeria rubra is the national flower of Nicaragua, where it is known under the local name "*sacuanjoche*". It is also the village flower of Asan-Maina on Guam.

Gallery

Close-up on flowers of a white variant

Close-up on flowers of a pink variant

Leaves sprouting

White *Plumeria rubra* tree

Pink *Plumeria rubra* tree

Red frangipani in Warje, India.

Plumeria rubra in Tahiti

Plumeria rubra in China

Frangipani rust (caused by the fungus *Coleosporium plumeriae*)

Closeup of *Plumeria rubra* in Jamshedpur

Plumeria rubra flowering when bare

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